

TRAINING A ROBUST IMAGE CLASSIFIER AGAINST ADVERSARIAL ATTACKS

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1 INTRODUCTION

Deep image classifiers are accurate on clean inputs yet can be fooled by *adversarial examples*: tiny, near-imperceptible pixel perturbations that flip the prediction. The task is to train a classifier on a 9-class, $3 \times 32 \times 32$ dataset (50,000 images) that keeps high accuracy on *both* clean and adversarially perturbed inputs. Submissions are scored by a unified metric, $\text{score} = 0.5 \cdot \text{clean acc} + 0.5 \cdot \text{robust acc}$, with a hard requirement of clean accuracy $> 50\%$. The evaluation attack is hidden, so the goal is *genuine* robustness rather than overfitting to one attack. A key constraint is that the server reconstructs a stock `torchvision` ResNet-18/34/50 and loads our `state_dict`; we may therefore not alter the architecture (no custom normalization layer or modified stem), and inputs arrive as pixels in $[0, 1]$.

2 APPROACH

Core method. Standard training gives $\approx 0\%$ robust accuracy, so we use *adversarial training* (AT): on every step we craft an L_∞ adversarial example with Projected Gradient Descent (PGD, Madry et al., 2018) and train on it. Throughout we use the standard CIFAR budget $\varepsilon = 8/255$, step size $\alpha = 2/255$ and 10 PGD steps; robust validation uses a stronger 20-step PGD on a held-out 4% split. We always save the checkpoint with the best validation score, which counters *robust overfitting* (robust accuracy peaks mid-training, then degrades).

Structural choices and why. Because the server rebuilds a plain ResNet, we train directly on $[0, 1]$ pixels (no normalization layer) so train- and test-time forward passes match, and clamp adversarial examples to $[0, 1]$. We add three ingredients that consistently helped: (i) *model capacity*: ResNet-50 over ResNet-18; (ii) a *cosine* learning-rate schedule with a 5-epoch warmup, which converges more smoothly than the step schedule; and (iii) *EMA weight averaging* (decay 0.999): we evaluate and submit the averaged weights, a near-free robustness gain. Optimisation is SGD (momentum 0.9, weight decay 5×10^{-4} , batch 192–256). Label smoothing (0.1) raised clean accuracy.

A diagnostic that set the strategy. Training ResNet-50 with *no* attack reached **0.81** clean accuracy. This showed the architecture does *not* cap clean accuracy; our adversarial models’ lower clean accuracy is purely the clean/robust trade-off, so the bottleneck for the score is *robustness*, not clean accuracy. We therefore focused later experiments on the robustness axis.

Methods explored. Beyond PGD-AT we implemented and tested TRADES (Zhang et al., 2019), Adversarial Weight Perturbation (AWP, Wu et al., 2020), and CutMix augmentation (Rebuffi et al., 2021). TRADES is numerically unstable at a high learning rate with the ResNet stem (we observed NaN losses); we stabilised it with LR warmup, a β warmup ($0 \rightarrow \beta$ over 10 epochs), gradient clipping (norm 1.0), and skipping non-finite batches.

3 RESULTS

Table 1 reports the best held-out validation score for each configuration. The recipe **ResNet-50 + PGD-AT + cosine + label smoothing + EMA** was the clear winner. Capacity (R18→R50) and the cosine/LS/EMA combination drove the main gains. TRADES traded clean for robust accuracy at a roughly constant sum; CutMix lowered clean accuracy too much; and AWP raised robustness slightly but cost more clean accuracy, so it did not improve the unified score on this dataset. These negative results help map the trade-off.

Configuration	Clean	Robust	Score
PGD-AT, ResNet-18 (baseline)	0.608	0.441	0.525
TRADES + EMA, ResNet-18	0.687	0.380	0.534
PGD-AT, ResNet-50	0.631	0.462	0.546
TRADES, ResNet-34 (cos+LS+EMA)	0.690	0.442	0.566
TRADES, ResNet-50 (cos+LS+EMA)	0.700	0.456	0.578
PGD-AT + AWP, ResNet-50	0.645	0.468	0.557
TRADES + AWP, ResNet-50	0.710	0.401	0.556
CutMix + PGD-AT, ResNet-50	0.660	0.421	0.541
PGD-AT, ResNet-50 (cos+LS+EMA)	0.705	0.486	0.596
Clean-only ceiling (no attack)	0.810	0.000	0.405

Table 1: Best validation results. “cos+LS+EMA” = cosine LR + label smoothing + EMA. All AT uses L_∞ , $\varepsilon = 8/255$.

Public leaderboard. Submissions tracked our local validation closely (server score $\approx$ local +0.005–0.01), so we could iterate without wasting the 60-minute-cooldown submissions. The leaderboard score climbed $0.537 \rightarrow 0.550 \rightarrow 0.604$ as we moved from ResNet-18/TRADES to the ResNet-50 recipe, with a best public score of **0.604**, placing in the top tier of the leaderboard.

4 CONCLUSION

We trained a genuinely robust ResNet-50 ($\approx 70\%$ clean, $\approx 49\%$ robust) using PGD adversarial training with a carefully tuned schedule, label smoothing, and EMA averaging. The central lesson is that robustness, not clean accuracy, is the binding constraint here, and that the standard robustness-boosting tricks each only shift the clean/robust trade-off rather than breaking it; closing the final gap would likely require generated/external data, the known route to higher robustness. Practically, adversarial vulnerability matters because the very systems we deploy in safety-critical settings (medical imaging, autonomous driving, content and fraud filters) can be fooled by perturbations invisible to humans, and an attacker need not know our model internals. Adversarial training is a partial but real defense: it buys meaningful robustness at a measurable cost in clean accuracy, and that trade-off, not perfect accuracy, is what responsible deployment must reason about.

All code to reproduce these results is available at <https://github.com/rimsha987/tml26-mia3>.

REFERENCES

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